

Physics Packet: Introducing Kinetic Friction Part 1

Name: _____

Date: _____

Part A: Reviewing Concepts — Box on a Level Surface (No Push Yet)

Consider a box sliding on a level surface.



1) Identify the forces:

a) _____.

b) _____.

2) Force directions:

a) Up _____.

b) Down _____.

3) If the box is not moving up or down, what can we say about the net force in the x and y -direction?

$$\sum F_x = \underline{\hspace{2cm}}$$

$$\sum F_y = \underline{\hspace{2cm}}$$

Part B: 1. Add a Push — Motion to the Right

Now a student pushes the box to the right and the box **slides** across the surface. If the surface is a frictionless surface, the box will keep going in the direction it is pushed in.



1) Identify the forces:

- a) _____.
- b) _____.
- c) _____.

2) Force directions:

- a) Up _____.
- b) Down _____.
- c) Right _____.

3) If the box is not moving up or down, what can we say about the net force in the x and y -direction?

$$\sum F_x = \underline{\hspace{2cm}}$$

$$\sum F_y = \underline{\hspace{2cm}}$$

Part B: 2. Add a Push — Motion to the Right

Now a student pushes the box to the right and the box **slides** across the surface. While it is sliding, kinetic friction acts.



1) Identify the forces:

- a) _____.
- b) _____.
- c) _____.
- d) _____.

2) Force directions:

- a) Up _____.
- b) Down _____.
- c) Left _____.
- d) Right _____.

3) If the box is not moving up or down, what can we say about the net force in the x and y -direction?

$$\sum F_x = \underline{\hspace{2cm}}$$

$$\sum F_y = \underline{\hspace{2cm}}$$

4) Formulas & Facts for Kinetic Friction

- a) Kinetic friction exists when the object is sliding/moving.

b) The magnitude of kinetic friction is:

$$f_k = \underline{\hspace{10cm}}$$

where μ_k is the coefficient of friction.

c) Since the surface is level and there is no vertical acceleration:

$$\sum F_y = 0 \quad \Rightarrow \quad F_N = \underline{\hspace{2cm}}$$

d) When the box is sliding right:

$$\sum F_x = F_{\text{push}} - f_k$$

and Newton's 2nd Law gives:

$$\sum F_x = ma$$

e) Is kinetic friction always “equal and opposite” to the push force?

Answer: _____

Practice Problems (Show Work)

Problem 1.

A 12 kg box slides on a level floor to the right. The coefficient of kinetic friction is $\mu_k = 0.25$.

1) Draw an Free Body Diagram.

2) What is F_g ?

3) What is F_N ?

4) What is f_k ?

5) What is $\sum F_x$?

6) What is $\sum F_y$?

Problem 2.

A 5 kg box slides on a level floor to the right. The coefficient of kinetic friction is $\mu_k = 0.31$.

1) Draw an Free Body Diagram.

2) What is F_g ?

3) What is F_N ?

4) What is f_k ?

5) What is $\sum F_x$?

6) What is $\sum F_y$?

Problem 3.

A 10 kg box is pushed to the right with $F_{\text{push}} = 40 \text{ N}$. The coefficient of kinetic friction is $\mu_k = 0.20$.

1) Draw an Free Body Diagram.

2) What is F_g ?

3) What is F_N ?

4) What is f_k ?

5) What is $\sum F_x$?

6) What is $\sum F_y$?

Problem 4.

A 25 kg box is pushed to the right with $F_{\text{push}} = 105 \text{ N}$. The coefficient of kinetic friction is $\mu_k = 0.25$.

1) Draw an Free Body Diagram.

2) What is F_g ?

3) What is F_N ?

4) What is f_k ?

5) What is $\sum F_x$?

6) What is $\sum F_y$?